

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

Course Code:	CSA6503	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 2 - Model Comparison Presentation (BERT vs GPT vs T5 vs LLaMA)
Weightage:	10% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
Student Name:	M.Someswar Reddy	Reg. No.:	192472081
Section Batch: /	CSE-AI	Date:	21/8/26

Instructions to Students

- **Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.**
- **Compare all four models: BERT, GPT, T5, and LLaMA.**
- Explain the **architecture and working principle** of each model.
- Compare their **pre-training objectives and capabilities**.
- Map each model to appropriate **real-world applications**.
- Identify the **strengths and limitations** of each model.
- Support the presentation using **credible research papers, official documentation, benchmarks, or other reliable sources**.
- Include at least one **comparison table or diagram**.
- Use relevant **real-world case studies or application examples**.
- Conclude by identifying the **most suitable model for the selected application** and justify the decision.
- Include proper **references/citations** in the presentation.

Code of Conduct

*I **M.SomeswarReddy/192472081** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M.Someswar Reddy

15. Compare BERT, GPT, T5, and LLaMA for AI chatbot development. Map each model to suitable chatbot applications.

AI CHATBOT DEVELOPMENT

Compare BERT, GPT, T5, and LLaMA for AI chatbot development. Map each model to suitable chatbot applications.

1. BERT

BERT is an encoder-only Transformer model mainly used for understanding text. It can be used in chatbots for intent detection, FAQ classification, question understanding, and sentiment analysis. BERT is not primarily designed for generating long conversational responses.

2. GPT

GPT is a decoder-only Transformer model designed for language generation. It can generate natural conversational responses and is suitable for customer-support chatbots, educational assistants, personal assistants, and general conversational AI.

3. T5

T5 is an encoder-decoder Transformer model that treats NLP problems as text-to-text tasks. It can be used for question answering, summarization, translation, and FAQ chatbot applications.

4. LLaMA

LLaMA is a decoder-only large language model designed for language generation. It can be used for educational chatbots, customer-support systems, coding assistants, and customizable conversational AI applications.

Comparison Table

Model	Architecture	Main Use	Chatbot Role	Limitation
BERT	Encoder-only	Text understanding	Intent detection, FAQ, classification	Limited text generation
GPT	Decoder-only	Text generation	Conversational chatbot	Can generate incorrect information
T5	Encoder-decoder	Text-to-text tasks	QA, summarization, FAQ	Less focused on open-ended conversation
LLaMA	Decoder-only	Text generation	Advanced conversational chatbot	Requires suitable computing resources

Suitable Chatbot Applications

BERT: FAQ systems, intent classification, customer-query routing.

GPT: customer support, personal assistants, educational chatbots.

T5: question answering, summarization, structured FAQ systems.

LLaMA: educational assistants, coding assistants, customized conversational systems.

Recommendation

For a general conversational chatbot, GPT or LLaMA is more suitable as the main response-generation model. BERT can be used for understanding and classification, while T5 is useful for question answering and other text-to-text tasks. A combined system can use BERT for intent detection and GPT or LLaMA for response generation.

Conclusion

BERT is mainly useful for understanding user input, GPT and LLaMA are strong choices for conversational text generation, and T5 is flexible for text-to-text tasks. The best choice depends on the chatbot's purpose.

36. A college wants to build a RAG-based document question-answering system using academic regulations and course documents. Compare BERT, GPT, T5, and LLaMA and recommend a suitable model.

RAG-BASED DOCUMENT QUESTION ANSWERING

A college wants to build a RAG-based document question-answering system using academic regulations and course documents. Compare BERT, GPT, T5, and LLaMA and recommend a suitable model.

1. What is RAG?

RAG stands for Retrieval-Augmented Generation. A RAG system first retrieves relevant information from documents and then uses a language model to generate an answer.

Basic RAG Flow

Student Question → Document Search/Retrieval → Relevant College Documents → Language Model → Final Answer

2. BERT in RAG

BERT can be used for understanding questions, semantic search, document retrieval, and classification. It is useful for finding relevant information from college regulations and course documents. It is not primarily designed for generating detailed conversational answers.

3. GPT in RAG

GPT can use retrieved information to generate natural-language answers. It is suitable for answer generation, summarization, conversational interaction, and follow-up questions.

4. T5 in RAG

T5 can use retrieved information as input and produce an answer as output. It is suitable for question answering, summarization, and other text-to-text tasks.

5. LLaMA in RAG

LLaMA can act as the generation model in a RAG pipeline. It can generate conversational answers based on retrieved college documents and can be customized for a college-specific system.

RAG Comparison Table

Model	Architecture	Retrieval	Answer Generation	RAG Role	Best Use
BERT	Encoder-only	Very suitable	Limited	Retriever/encoder	Finding relevant information
GPT	Decoder-only	Possible	Excellent	Generator	Natural-language answers
T5	Encoder-decoder	Possible	Good	Generator	QA and text transformation
LLaMA	Decoder-only	Possible	Excellent	Generator	Custom conversational RAG

Recommended Architecture

College Documents → Retriever/Embedding Model → Relevant Documents → LLaMA → Final Answer → Student

Recommendation

For the college RAG system, a combination of BERT and LLaMA can be recommended. BERT can help retrieve or understand relevant information, while LLaMA can generate a natural conversational answer using the retrieved information.

Conclusion

In a RAG-based college question-answering system, BERT is useful for retrieval and understanding, while GPT, T5, and LLaMA can be used for answer generation. A BERT plus LLaMA pipeline provides a clear separation between finding relevant information and generating the final response.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Architectural Comparison Accuracy	30%	Accurate, in-depth comparison of architectures and use-cases	Accurate comparison with minor gaps	Superficial comparison of models	Inaccurate or confused comparison
Application Mapping	25%	Clearly maps each model to appropriate real-world applications	Reasonable mapping to applications	Weak mapping to applications	No meaningful mapping presented
Research Depth	20%	Well-researched with credible sources/benchmarks cited	Adequately researched	Limited research depth	Unresearched / opinion-based content
Delivery	15%	Confident, engaging, well-paced delivery	Clear delivery with minor pacing issues	Hesitant delivery	Unclear or unprepared delivery
Slide Design & References	10%	Well-designed slides with proper references	Adequate slides, minor reference gaps	Basic slides, poor referencing	No slides or references provided

Marks Evaluation:

Question No.	Architectural Comparison Accuracy (30%)	Application Mapping (25%)	Research Depth (20%)	Delivery (15%)	Slide Design & References (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						